FYDP-I Individual Journal (Week 1)

Project Title: Benchmarking Large Language Models for Bangladeshi Statutory Law: A Study on Accuracy and Hallucination in Bangla
Course Code: CSE 4000A (FYDP I)

1. Mapping with Complex Engineering Problems (P1–P7) and Complex Engineering Activities (A1–A5)

1.1 Complex Engineering Problem Attributes

- **P1 – Depth of Knowledge Required (Mandatory):** Requires knowledge of Natural Language Processing (NLP), Large Language Models (LLMs), Bangla language processing, legal text analysis, and AI evaluation.
- **P4 – Familiarity of Issues:** *No standardized native Bangla MCQ benchmark dataset exists for codified Bangladeshi statutory law.*
- **P6 – Extent of Stakeholder Involvement and Conflicting Requirements:** Balances the growing demand for automated legal information with the need to minimize LLM hallucinations, as inaccurate legal responses may mislead citizens and lead to legal or financial consequences.

1.2 Complex Engineering Activity Attributes

- **A1 – Range of Resources:** Integrates diverse technologies, including PDF parsers (PyMuPDF), optical character recognition engines (Tesseract OCR), data validation frameworks, and various open-source/proprietary LLM APIs.
- **A2 – Level of Interaction:** Requires resolving interacting technical challenges across text extraction, OCR correction, subword tokenization limits, dataset validation, and LLM evaluation.
- **A3 – Innovation:** Involves building a novel, domain-specific statutory dataset pipeline from scratch and establishing a new three-dimensional legal hallucination evaluation framework.

2. Problem Classification: Complex vs. Complicated Engineering Problem

This project qualifies as a **Complex Engineering Problem**.

Unlike a complicated engineering problem, which can be solved using standard methods and linear workflows, this project combines Bangla Natural Language Processing (NLP), legal text processing, benchmark dataset construction, and Large Language Model (LLM) evaluation. It requires multiple technical decisions and engineering judgment, as there is no standard procedure that guarantees the best solution. Furthermore, the probabilistic nature of LLM hallucinations and the need for high legal accuracy make the problem complex, since incorrect legal responses may lead to real-world legal or financial consequences.

3. Absence of a Single Exact/Fixed Solution

The project lacks a single, fixed solution due to the following reasons:

- **Model-Dependent Variability:** Different LLMs (e.g., GPT-4o, Gemini, Llama, Qwen) produce different outputs for the same legal query, meaning no single model can be treated as a universal ground-truth solution.
- **Open-Ended Frameworks:** Different prompting strategies (zero-shot, few-shot, and chain-of-thought) and evaluation metrics produce different results, rather than converging on a single optimal approach.
- **No Universal Benchmarking Framework:** There is no single benchmarking framework that works best for every model, since evaluation criteria must be adapted depending on the model architecture and task design.
- **Probabilistic Core of AI:** Because generative models operate on probabilistic token weights, absolute zero hallucination cannot be hardcoded; it requires continuous, multi-faceted optimization.

4. Realistic Availability of Data, Tools, and Equipment

All necessary resources are realistically available to ensure project feasibility:

- **Data:** The target text (Bangladesh Road Transport Act, 2018) is a publicly accessible 38-page government PDF. An initial benchmark of 299 expert-reviewed MCQ pairs has already been successfully compiled and open-sourced.
- **Tools:** The text processing and data validation tools (Python, Pandas, PyMuPDF, Tesseract OCR) are entirely open-source and free to use.
- **Equipment and Models:** Public foundation models (e.g., Llama and Qwen) are openly accessible via Hugging Face for local deployment, and proprietary models (GPT-4o and Gemini) are available through standard, low-cost developer APIs.